Quality Control (QC) Audit Report

Software Requirements Specification (SRS)

Biometric Student Attendance System

Veridion Labs

March 19, 2026

Team Members	Department	Role	Employee ID
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Table 1: Veridion Labs QC Audit Team

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Note on Hyperlinks: Throughout this document, all hyperlinks are displayed in blue and underlined. These links are fully clickable in the PDF viewer and will navigate to the relevant section, finding, or appendix.

1 Executive Summary

This report presents the results of a Quality Control (QC) validation of the Software Requirements Specification (SRS) for the **Biometric Student Attendance System**, prepared by Nevon Solutions for BioSec Educational Institution (Document ID: NS-SRS-2026-001, Version 0.8, dated 3 March 2026). The objective of this QC audit is to identify defects that remain in the SRS after the completion of Eliciting, Analysing, and Specifying.

The audit focused on three principal categories of SRS defects expected at the Validating stage:

- **Authorship defects** – where intended meanings are not expressed clearly enough.
- **Omission defects** – where required scenarios, conditions, or supporting content are missing.
- **Contradiction defects** – where different parts of the SRS imply inconsistent behaviour.

The inspection found that the SRS is generally well-structured and covers most core functional areas including biometric attendance recording, leave management, compliance monitoring, and push notifications. However, a number of defects were detected including a contradiction in the scope of two-factor authentication enforcement, an inconsistency in attendance status terminology across sections, missing requirements for biometric enrolment, password recovery, offline limit handling, and session scheduling, as well as grammatical and editorial errors. The SRS sign-off section also contains blank approver signatures.

These findings indicate that the SRS is **conditionally acceptable**, subject to BioSec's review of the identified defects. No recommendations for correction are provided; the customer will determine any necessary actions.

2 Introduction

2.1 Purpose

This Quality Control (QC) Audit Report presents the findings of an independent inspection of the Software Requirements Specification (SRS) prepared for the **Biometric Student Attendance System** (NS-SRS-2026-001). The purpose of this QC activity is to identify defects remaining in the SRS work product before the project progresses into downstream stages such as system design and implementation.

In accordance with the validating stage of Requirements Engineering, QC focuses on defect detection in the **requirements work product itself**. The QC team records findings but does not propose fixes; the customer determines any corrective actions.

2.2 Document Conventions

This document is an audit report rather than a replacement SRS. It records detected defects and explains their implications.

Audit findings use the format `QC-FND-XXX`.

Error detection instances use the format `QC-ERR-XXX`.

2.3 Intended Audience and Reading Suggestions

This report is intended for the BioSec Educational Institution requirements engineering team, project management personnel, and relevant stakeholders involved in SRS review and approval.

Readers should begin with the Executive Summary and Sections 1–3, then proceed to the QC Checklist, Error Detection, and detailed Findings.

2.4 Audit Scope

The audit covers the SRS (NS-SRS-2026-001) as a QC exercise. The scope includes:

- detection of authorship defects (ambiguity, vagueness, grammatical errors)
- detection of latent contradictions across sections
- detection of omission of topics (missing scenarios, exception paths, uncovered capabilities)
- checking codification and traceability coherence
- checking structural, editorial, and presentation consistency

3 Audit Basis

3.1 Basis and Nature of QC

This audit assumes the SRS is mature after earlier RE stages. QC inspects the work product itself, identifying defects without direct amendment. The audit team reports findings for customer consideration.

3.2 Kinds of Errors Pursued

- **Authorship:** unclear expression of intended meaning.
- **Latent contradictions:** conflicting requirements across sections.
- **Omission:** missing topics, edge conditions, or supporting information.
- **Traceability:** incomplete codification, source mapping, or cross-references.
- **Editorial:** formatting, naming, or presentation inconsistencies.

3.3 Severity Classification

- **High:** significantly affects interpretation, implementation, or expected behaviour.
- **Medium:** introduces ambiguity without changing core intent.
- **Low:** minor editorial issues with no material impact.

3.4 Documents Reviewed

- SRS: Biometric Student Attendance System (NS-SRS-2026-001, Version 0.8, 3 March 2026)
- BioSec Project Specification (BS-PS-2026-001, Version 1.0)
- Requirements models: Context DFD, Wireframes, Use Case Diagram, ERD (Figures 1–4)
- Inceptive Requirements Traceability Matrix (iRTM, Appendix D)

- Change Mitigation Strategies (Section 8)
- Data Dictionary (Appendix C) and Glossary (Appendix A)

4 QC Plan

4.1 Target Workproduct

The **Software Requirements Specification (SRS)** for the Biometric Student Attendance System, prepared by Nevon Solutions for BioSec Educational Institution (NS-SRS-2026-001, Version 0.8).

4.2 Context and Goals

The SRS serves as the contractual baseline for all downstream design, development, testing, and delivery activities. This audit verifies the SRS's clarity, completeness, and internal consistency, documenting any detected defects for BioSec's consideration and action.

4.3 Access and Audit Conditions

The QC team reviewed the released SRS document and all supporting artefacts provided for audit, including diagrams, appendices, the iRTM, and change mitigation strategies.

4.4 Inspection Techniques Used

1. Close document inspection
2. Perspective-based reading (developer, tester, PM, compliance officer)
3. Cross-section contradiction checking
4. Codification and traceability verification
5. Model-to-text consistency inspection
6. Omission mapping against expected SRS topics
7. Scenario-based reading (biometric attendance and leave management scenarios)
8. Assumption reading
9. Double-checking dates, facts, figures, and identifiers
10. Spellcheck and grammar review

4.5 Work Breakdown

Team roles and responsibilities are detailed in Appendix A, Table A-1: QC Audit Team Work Allocation.

4.6 Indicative Timeline

The audit schedule is shown in Appendix A, Table A-2: Indicative QC Audit Timeline.

4.7 Reporting

The QC team records findings in this report; the audit team does not amend the SRS.

5 QC Checklist

The following checklist was used by the QC audit team to inspect the SRS work product. The checklist items represent critical aspects of a high-quality SRS, derived from industry standards and lecture guidelines, and are assessed against the Biometric Student Attendance System SRS specifically.

For readability in the main report, the full checklist register is presented in Appendix B, Table B-1: QC Inspection Checklist. The appendix table records each checklist item together with the inspection result applied during the audit.

6 Error Detection

The QC audit identified instances of three principal QC error types: authorship, omission, and contradiction. Each detected instance was recorded with its location in the SRS and the characteristic of the defect observed during inspection.

For readability in the main report, the consolidated error detection register is presented in Appendix C, Table C-1: Detected QC Error Instances. The appendix table serves as the detailed supporting record for the findings discussed in Section 7. A summary of all findings is separately provided in Appendix D, Table D-1: QC Findings Summary.

7 QC Findings

7.1 Finding Summary

The audit identified 12 findings across three defect types. A consolidated summary of all findings is provided in Appendix D, Table D-1: QC Findings Summary. Detailed analysis for each finding is presented in Section 7.2 below.

7.2 Detailed Findings

QC-FND-001: Contradiction in Two-Factor Authentication Scope

Two conflicting requirements exist on which user classes must use 2FA. UR-MOB-06 mandates 2FA for parents and teachers on the mobile app, while NFR-SEC-03 restricts 2FA only to administrators on the web portal. See Appendix D, Table D-1: QC Findings Summary for the finding summary entry.
Related error detection: QC-ERR-001
Type: Contradiction
Severity: High

Context. Section 3.1 UR-MOB-06 requires a 2-Factor Authentication screen process for "parent and teacher" mobile application users. Section 5.3 NFR-SEC-03 separately mandates that the system "shall enforce two-factor authentication (2FA) for all administrator accounts accessing the web portal" – with no mention of parents or teachers.

Result. A developer cannot determine whether 2FA is required for all three user classes or only administrators. Implementing 2FA for parents and teachers incurs additional development effort and user experience considerations not reflected in the NFR.

Analysis. These two requirements are mutually inconsistent on the scope of 2FA enforcement. One must be incorrect or the SRS must be clarified to specify a single authoritative position on which user classes require 2FA.

QC-FND-002: Contradiction in Attendance Status Terminology

The attendance status value “Attended” used in the UI requirements conflicts with “Present” used in the AttendanceRecord data dictionary ENUM, creating a direct inconsistency between the presentation layer and the data layer. See Appendix D, Table D-1: QC Findings Summary for the finding summary entry.
Related error detection: QC-ERR-002
Type: Contradiction
Severity: High

Context. Section 3.1 UR-MOB-01 states attendance status is shown as “Attended, Absent, Late, or Excused”. Section 4.3.2 Stimulus/Response Sequences states the system displays records with status “Excused, Attended, Late, Absent”. However, Appendix C Table C.4 AttendanceRecord defines the status ENUM as “Present, Late, Absent, Excused” – using “Present” rather than “Attended”.

Result. The database schema uses a different status value from the UI and functional requirements. This will cause implementation inconsistency between the data layer and presentation layer unless one authoritative value is confirmed.

Analysis. The term “Attended” in UI requirements directly conflicts with “Present” in the data dictionary. This is a latent contradiction that will surface during development and testing.

QC-FND-003: Contradiction in Report Generation Time Specification

The stimulus/response description for report generation omits the query-type distinction that NFR-PERF-03 explicitly requires, creating an inconsistency between functional description and non-functional measurement. See Appendix D, Table D-1: QC Findings Summary for the finding summary entry.
Related error detection: QC-ERR-003
Type: Contradiction
Severity: Medium

Context. Section 4.6.2 Stimulus/Response Sequences states that “the administrator generates an attendance report, which the system returns within 10–60 seconds depending on data volume”, without distinguishing query types. NFR-PERF-03 specifies “within 10 seconds for single class queries and within 60 seconds for institution-wide reports”.

Result. The stimulus/response description omits the distinction between single-class and institution-wide queries, making the behaviour ambiguous for testers and architects relying on that section.

Analysis. While NFR-PERF-03 is precise, the omission of the distinction in Section 4.6.2 creates an inconsistency between how the feature is described functionally and how it is measured non-functionally.

QC-FND-004: Omission of Biometric Enrolment Process

The SRS specifies extensive biometric scanning behaviour but is entirely silent on the student fingerprint enrolment process – a prerequisite capability without which the attendance system cannot function. See Appendix D, Table D-1: QC Findings Summary for the finding summary entry.
Related error detection: QC-ERR-004
Type: Omission
Severity: High

Context. The SRS extensively describes biometric fingerprint scanning for attendance recording across Sections 3.2, 4, and Appendix C. However, there is no requirement anywhere in the SRS defining how students are initially enrolled into the biometric system – the fingerprint registration process, who performs it, under what conditions, and what happens if enrolment fails or is rejected.

Result. Without enrolment requirements, the development team has no specification for a core prerequisite of the entire biometric attendance function. The system cannot operate without students being enrolled, yet the SRS is silent on this process.

Analysis. This is a significant functional omission. The biometric attendance system is entirely dependent on prior enrolment, making this a high-severity gap in the requirements baseline.

QC-FND-005: Omission of Password Reset and Account Recovery

The SRS specifies account lockout after failed login attempts but provides no requirement for how locked-out users can recover access, leaving a critical gap in the authentication lifecycle. See Appendix D, Table D-1: QC Findings Summary for the finding summary entry.
Related error detection: QC-ERR-005
Type: Omission
Severity: High

Context. NFR-SEC-04 specifies that the system shall lock user accounts after a configurable number of failed login attempts. Section 4.2.2 Stimulus/Response confirms that after five unsuccessful attempts, the account is locked. However, no requirement in the SRS defines how a user recovers access – there is no password reset, account unlock, or self-service recovery flow specified anywhere.

Result. Users who trigger the lockout policy have no defined path to regain access. Administrators, parents, and teachers could be permanently locked out with no recovery mechanism, directly impacting system usability and operational continuity.

Analysis. Account lockout without a corresponding recovery mechanism is an incomplete functional capability. This omission will require resolution before the system can be deployed to real users.

QC-FND-006: Omission of Offline Limit Exceeded Handling

The SRS defines a 48-hour offline mode but contains no requirement for system behaviour when this limit is exceeded or for resolving data conflicts upon reconnection. See Appendix D, Table D-1: QC Findings Summary for the finding summary entry.
Related error detection: QC-ERR-006
Type: Omission
Severity: High

Context. Section 2.5 specifies that the system must support offline attendance viewing and leave request drafting for up to 48 hours, with a 50MB local storage limit. NFR-SEC-09 confirms cached data is auto-purged after 48 hours. However, there is no requirement specifying what happens when the 48-hour offline limit is exceeded mid-session, or how data synchronisation conflicts are resolved when connectivity is restored after an offline period.

Result. If a user exceeds the 48-hour offline limit, the system behaviour is undefined. Similarly, if locally drafted leave requests conflict with server-side data upon reconnection, there is no specified resolution strategy.

Analysis. Offline mode is an explicitly scoped capability. The absence of requirements for limit-exceeded behaviour and conflict resolution leaves developers to make assumptions that may be inconsistent with stakeholder expectations.

QC-FND-007: Omission of Session Creation and Management Requirements

Academic sessions are referenced extensively as a prerequisite for attendance capture, but no requirement in the SRS defines how sessions are created, scheduled, or managed within the system.

See Appendix D, Table D-1: QC Findings Summary for the finding summary entry.
Related error detection: QC-ERR-007
Type: Omission
Severity: Medium

Context. BR-02 states that attendance capture is only permitted during "officially scheduled academic sessions" and that scans outside session windows are rejected. FR-ATT-03 references a "session_id" as a stored field in AttendanceRecord. The AttendanceRecord data dictionary (Appendix C, Table C.4) also references a "session_id FK to scheduled academic session". However, no requirement anywhere in the SRS defines how academic sessions are created, configured, scheduled, or managed within the system.

Result. The system cannot enforce BR-02 or populate session_id without sessions being defined in the system. The SRS assumes sessions exist but never specifies how they come to exist.

Analysis. This is a functional gap. Session management is a prerequisite for the attendance capture business rule to function correctly.

QC-FND-008: Ambiguous Use of "Periodic Synchronisation"

The product functions summary uses the vague term "periodic synchronisation" without specifying a frequency, inconsistent with the precise 5-minute figure stated in FR-COM-01. See Appendix D, Table D-1: QC Findings Summary for the finding summary entry.
Related error detection: QC-ERR-008
Type: Authorship
Severity: Medium

Context. Section 2.2 Product Functions describes attendance data access as providing "role-scoped mobile access for parents (child only) and teachers (assigned classes only), with periodic synchronisation". FR-COM-01 specifies that mobile applications shall synchronise attendance data with the backend "every 5 minutes when connected".

Result. The word "periodic" in Section 2.2 is vague and does not communicate the 5-minute interval. A reader relying on the product functions summary alone would not know the synchronisation frequency.

Analysis. While FR-COM-01 is precise, the inconsistency between vague language in the product summary and specific language in the functional requirement reduces overall document clarity.

QC-FND-009: Grammatical Error in QA-USAB-01

A double verb construction in QA-USAB-01 – "shall allow ... must be able to" – is grammatically incorrect and introduces ambiguity about the nature of the obligation. See Appendix D, Table D-1: QC Findings Summary for the finding summary entry.
Related error detection: QC-ERR-009
Type: Authorship
Severity: Low

Context. Section 5.4.1 QA-USAB-01 states: "The system shall allow parent and teacher users must be able to complete primary tasks ... within three interactions from the home screen".

Result. The double verb construction "shall allow ... must be able to" is grammatically incorrect and creates ambiguity about the nature of the obligation – whether the system is the subject of the requirement or the users are.

Analysis. While the intended meaning is reasonably clear from context, the grammatical error introduces imprecision in a quality attribute requirement that should be unambiguous for verification purposes.

QC-FND-010: Capitalisation Inconsistency in UR-WEB-08 and UR-WEB-09

Requirements UR-WEB-08 and UR-WEB-09 begin with a lowercase "the portal shall" while all other requirements in Section 3.1 consistently use the capitalised "The portal shall", indicating an editorial oversight. See Appendix D, Table D-1: QC Findings Summary for the finding summary entry.
Related error detection: QC-ERR-010
Type: Authorship
Severity: Low

Context. Section 3.1 requirements UR-WEB-08 and UR-WEB-09 begin with lowercase: "the portal shall display an activity log..." and "the portal shall include a 2-Factor Authentication screen...". All other interface requirements in the same section (UR-GUI-01 through UR-WEB-07 and UR-WEB-10) begin with a capitalised "The".

Result. The inconsistency reduces document professionalism and suggests an editorial oversight in the final review of Section 3.1.

Analysis. While this does not affect the meaning of the requirements, it is indicative of an insufficient final editorial pass prior to SRS release.

QC-FND-011: Unsigned SRS Sign-off Section

All four designated approvers in the SRS sign-off section – two from Nevon Solutions and two from BioSec – have entirely blank signature and date fields, indicating the SRS was released without their formal approval. See Appendix D, Table D-1: QC Findings Summary for the finding summary entry.
Related error detection: QC-ERR-011
Type: Authorship
Severity: Low

Context. Section 9 SRS Sign-off contains a Designated Approvers table for both Vendor (Nevon Solutions: James Koh – Technical Lead, Rachel Lee – System Architect Lead) and Customer (BioSec: Marcus Tan Wei Liang – Project Manager, Daniel Rodriguez – Head of Operations). All four approver rows have entirely blank Signature and Date fields.

Result. The SRS was released without designated approver sign-off. The document's own sign-off certifies that it "constitutes the approved development baseline", but the approver table contradicts this by showing no approvals.

Analysis. While the Project Lead (Loh Wen Xuan) has signed the Vendor Authenticator section, the absence of designated approver signatures raises a question about whether the SRS was formally reviewed and approved by all required parties before release.

QC-FND-012: RBAC Not Defined in Glossary

RBAC is one of the most frequently referenced acronyms in the SRS but is absent from Appendix A Glossary, which defines 27 other terms including abbreviations of similar nature. See Appendix D, Table D-1: QC Findings Summary for the finding summary entry.
Related error detection: QC-ERR-012
Type: Omission
Severity: Medium

Context. The acronym RBAC (Role-Based Access Control) is used throughout the SRS – in Section 4.1 heading, FR-RBAC-01, QA-INTEG-01, and multiple other locations. Appendix A Glossary defines 27 terms including FR, NFR, BOR, TR, and other abbreviations, but RBAC is not among them.

Result. A reader unfamiliar with RBAC would not find a definition in the SRS itself despite RBAC being central to the system's security architecture.

Analysis. Given that RBAC is one of the most frequently referenced concepts in the SRS and the Glossary explicitly defines other abbreviations of similar nature, the omission of RBAC is an inconsistency in the addenda.

8 Conclusion

The QC audit concludes that the Software Requirements Specification for the Biometric Student Attendance System is substantively well-structured and reflects considerable work carried out during the earlier Requirements Engineering stages. The SRS demonstrates thorough coverage of functional areas including biometric scanning, leave management, compliance monitoring, notification delivery, and traceability.

However, the document contains defects typical of the validating stage. Two high-severity contradictions were identified – the scope of 2FA enforcement and the inconsistency in attendance status terminology between the UI requirements and the data dictionary – both of which will directly impact implementation correctness. Three high-severity omissions were found concerning the biometric enrolment process, the absence of a password recovery mechanism, and undefined behaviour at the offline mode limit. Additional medium and low severity authorship and omission defects were also detected.

The QC team does not directly modify the SRS. Instead, the findings documented in this report are presented for BioSec's consideration. The customer will determine what actions, if any, should be taken to address the identified defects.

Overall, the SRS is assessed as **conditionally acceptable**, pending customer review and resolution of the findings.

9 Sign-off

9.1 QC Audit Team Authentication

Name	Role	Authentication
Muhammad Ridwan Putra Jasni	QC Audit Lead	
Lin Yuhao	QC Auditor	
Muhammad Mikhail Bin Mazlan	QC Auditor	
Kannan s/o Rajamohan	QC Auditor	

Table 2: QC Audit Team Sign-Off

10 Addenda

10.1 Glossary

Term	**Definition**
QC	Quality Control – independent inspection of a work product to detect defects, in this case the SRS.
SRS	Software Requirements Specification – the requirements baseline document for downstream development.
Authorship defect	Defect caused by unclear, imprecise, or grammatically incorrect expression of an intended requirement meaning.
Omission defect	Defect in which required content, scenario, capability, or supporting information is absent from the SRS.
Contradiction defect	Defect in which different parts of the SRS imply inconsistent or conflicting system behaviour.
Traceability coherence	The degree to which requirements, models, sources, and cross-references link together clearly and consistently throughout the SRS.
2FA	Two-Factor Authentication – a security mechanism requiring two forms of verification before granting access.
RBAC	Role-Based Access Control – a method of restricting system access based on the roles assigned to users.
PDPA	Personal Data Protection Act 2012 (Singapore) – the legislation governing collection, use, and disclosure of personal data.
iRTM	Inceptive Requirements Traceability Matrix – a matrix tracing PS requirements to SRS requirements.

Table G-1: QC Report Glossary

Appendix A: QC Plan Tables

A.1 QC Audit Team Work Allocation

Role	Assigned Member	Primary Responsibility
QC Audit Lead	Muhammad Ridwan Putra Jasni	Coordinate audit activities, consolidate findings, ensure consistency of final report, apply cross-section contradiction checking and assumption reading.
Requirements Reviewer	Lin Yuhao	Inspect functional and non-functional requirement sections for ambiguity, contradiction, and omission; perform perspective-based reading (developer, tester) and scenario-based reading for biometric attendance and leave management flows.
Traceability Reviewer	Muhammad Mikhail Bin Mazlan	Review iRTM references, source mappings, data dictionary, and section-to-section coherence; double-check all identifiers, cross-references, and figure citations.
Editorial and Structure Reviewer	Kannan s/o Rajamohan	Review document structure, section naming, formatting consistency, and overall presentation quality; conduct spellcheck and grammar review, verify figures, tables, and sign-off completeness.

Table A-1: QC Audit Team Work Allocation

A.2 Indicative QC Audit Timeline

Stage	Activity	Indicative Duration
Stage 1	Individual inspection of assigned SRS sections using selected techniques (close reading, perspective-based, scenario-based)	3 hours
Stage 2	Consolidation meeting and de-duplication of identified findings	1 hour
Stage 3	Severity review and documentation of findings	1 hour
Stage 4	Final report formatting, checking, and team sign-off	1 hour

Table A-2: Indicative QC Audit Timeline

Appendix B: QC Checklist

Item	Focus	QC Checklist Item	Inspection Result
1. Introduction Section			
QC1	Correctness	Does the Introduction explicitly define the system purpose, target user base, and operational boundaries (in-scope vs. out-of-scope)?	Satisfied
QC2	Completeness	Does the Introduction identify all intended audiences and provide reading guidance for each?	Satisfied
QC3	Consistency	Is the system name, document ID, version number, and project title consistent across the title page, header, and body?	Satisfied
2. Glossary / Definitions Section			
QC4	Completeness	Are all domain-specific acronyms and abbreviations used in the SRS defined in the Glossary?	Partially satisfied (RBAC missing)
QC5	Relevance	Does every term defined in the Glossary appear at least once in the SRS body?	Satisfied
QC6	Consistency	Is each system role or entity referred to by a consistent term throughout the document?	Partially satisfied
3. Overall Description Section			
QC7	Completeness	Are all user classes identified with their technical expertise, access privileges, and responsibilities?	Satisfied
QC8	Completeness	Does the SRS specify the intended operational environment (OS, platform, hardware, network)?	Satisfied
QC9	Correctness	Are product function descriptions measurably correct and consistent with detailed requirements in later sections?	Partially satisfied
4. Functional Requirements Section			
QC10	Correctness	Does each functional requirement describe a single, testable behaviour using precise language?	Partially satisfied
QC11	Completeness	Does the SRS include requirements for all critical system processes, including biometric enrolment?	Not satisfied
QC12	Consistency	Is every requirement assigned a unique identifier following a consistent naming convention?	Satisfied
5. Interface Requirements Section			
QC13	Completeness	Does the Interface section describe purpose, data exchanged, and communication method for each external interface?	Satisfied
QC14	Correctness	Are trigger conditions, expected data payloads, and error handling defined for each external interface?	Partially satisfied
QC15	Consistency	Do all interface requirements use consistent capitalisation and grammatical structure?	Not satisfied
6. Non-Functional Requirements Section			
QC16	Measurability	Are all NFRs quantified with measurable, testable limits (e.g., response time, availability percentage)?	Partially satisfied
QC17	Completeness	Does the SRS address all relevant NFR categories (performance, security, safety, reliability, usability, maintainability)?	Satisfied

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Item	Focus	QC Checklist Item	Inspection Result
QC18	Consistency	Are NFR obligations internally consistent – e.g., 2FA scope consistent across UR-MOB-06 and NFR-SEC-03?	Not satisfied
7. Addenda / Supporting Artefacts			
QC19	Traceability	Does the iRTM map all PS requirements to corresponding SRS requirements with correct cardinality?	Satisfied
QC20	Correctness	Does every diagram include a descriptive title, revision note, and standard notation accurately representing the system?	Satisfied
QC21	Completeness	Is the SRS sign-off section fully completed with signatures and dates from all designated approvers?	Not satisfied

Table B-1: QC Inspection Checklist

Appendix C: Error Detection Table

Error ID	Error Type	Location	Characteristic of Error
QC-ERR-001	Contradiction	Section 3.1 UR-MOB-06 vs Section 5.3 NFR-SEC-03	UR-MOB-06 requires a 2-Factor Authentication screen for "parent and teacher" mobile app users. NFR-SEC-03 enforces 2FA only for "all administrator accounts accessing the web portal". These statements directly conflict on which user classes require 2FA.
QC-ERR-002	Contradiction	Section 3.1 UR-MOB-01 vs Section 4.3.2 vs Appendix C Table C.4	UR-MOB-01 and Section 4.3.2 use the status value "Attended", while the AttendanceRecord data dictionary (Table C.4) defines the status ENUM as "Present, Late, Absent, Excused" – using "Present" instead of "Attended". The UI layer and data layer use different values for the same status.
QC-ERR-003	Contradiction	Section 4.6.2 Stimulus/Response Sequences vs NFR-PERF-03	Section 4.6.2 states reports are returned "within 10–60 seconds depending on data volume" without distinguishing between query types. NFR-PERF-03 specifies 10 seconds for single class queries and 60 seconds for institution-wide reports. The omission of this distinction in Section 4.6.2 creates an inconsistency with the NFR.
QC-ERR-004	Omission	Entire SRS – Sections 3.2, 4, Appendix C	The SRS specifies extensive biometric scanning behaviour but contains no requirement for the student biometric enrolment process – how fingerprints are initially registered, who performs the enrolment, what constitutes a successful enrolment, or how enrolment failures are handled.
QC-ERR-005	Omission	Section 4.2, NFR-SEC-04	NFR-SEC-04 specifies account lockout after failed login attempts, and Section 4.2.2 confirms lockout after five failures. However, no requirement defines a password reset, account unlock, or self-service recovery process. Users who trigger lockout have no defined path to regain access.
QC-ERR-006	Omission	Section 2.5, NFR-SEC-09	Section 2.5 and NFR-SEC-09 specify a 48-hour offline mode with auto-purge of cached data. No requirement specifies what happens when this limit is exceeded mid-session, nor how data sync conflicts are resolved when connectivity is restored after an offline period.

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Error ID	Error Type	Location	Characteristic of Error
QC-ERR-007	Omission	BR-02, FR-ATT-03, Appendix C Table C.4	BR-02 states attendance capture is only permitted during “officially scheduled academic sessions”. FR-ATT-03 and the AttendanceRecord data dictionary reference a session_id. However, no requirement defines how academic sessions are created, scheduled, or managed in the system.
QC-ERR-008	Authorship	Section 2.2 Product Functions vs FR-COM-01	Section 2.2 describes attendance data access with “periodic synchronisation” – a vague term with no frequency specified. FR-COM-01 specifies synchronisation every 5 minutes when connected. The imprecise language in Section 2.2 is inconsistent with the specific figure in FR-COM-01.
QC-ERR-009	Authorship	Section 5.4.1 QA-USAB-01	QA-USAB-01 states “The system shall allow parent and teacher users must be able to complete primary tasks . . . within three interactions”. The double verb construction “shall allow . . . must be able to” is grammatically incorrect, creating ambiguity about the nature of the obligation.
QC-ERR-010	Authorship	Section 3.1 UR-WEB-08 and UR-WEB-09	UR-WEB-08 begins “the portal shall display an activity log. . . ” and UR-WEB-09 begins “the portal shall include a 2-Factor Authentication screen. . . ” – both use lowercase “the”. All other requirements in Section 3.1 (UR-GUI-01 through UR-WEB-07, UR-WEB-10) begin with the capitalised “The”.
QC-ERR-011	Authorship	Section 9 SRS Sign-off	The Designated Approvers table lists four approvers – James Koh (Technical Lead), Rachel Lee (System Architect Lead), Marcus Tan Wei Liang (BioSec Project Manager), and Daniel Rodriguez (BioSec Head of Operations) – all with entirely blank Signature and Date fields, indicating the SRS was released without their formal approval.
QC-ERR-012	Omission	Appendix A Glossary vs Sections 4.1, FR-RBAC-01, QA-INTEG-01	The acronym RBAC is used extensively throughout the SRS including in the Section 4.1 heading, FR-RBAC-01, and QA-INTEG-01. Appendix A Glossary defines 27 terms but does not include RBAC, leaving the acronym undefined within the SRS.

Table C-1: Detected QC Error Instances

Appendix D: QC Findings Summary

Finding ID	Type	Severity	Related Error	Short Description
QC-FND-001	Contradiction	High	QC-ERR-001	2FA required for parents and teachers in UI requirements but only for administrators in security NFRs.
QC-FND-002	Contradiction	High	QC-ERR-002	Attendance status terminology inconsistent – "Attended" vs "Present" used across sections and data dictionary.
QC-FND-003	Contradiction	Medium	QC-ERR-003	Report generation time stated without query-type distinction in stimulus/response, contradicting NFR-PERF-03.
QC-FND-004	Omission	High	QC-ERR-004	No requirement for biometric enrolment process – how students are initially registered in the system.
QC-FND-005	Omission	High	QC-ERR-005	Account lockout specified but no password reset or account recovery flow requirement exists.
QC-FND-006	Omission	High	QC-ERR-006	Offline mode limit specified but no requirement for what happens when limit is exceeded or sync conflicts arise.
QC-FND-007	Omission	Medium	QC-ERR-007	Sessions referenced throughout but no requirement defining how sessions are created or managed.
QC-FND-008	Authorship	Medium	QC-ERR-008	"Periodic synchronisation" in Section 2.2 is vague and inconsistent with the specific 5-minute figure in FR-COM-01.
QC-FND-009	Authorship	Low	QC-ERR-009	Grammatical error in QA-USAB-01 – double verb construction "shall allow … must be able to".
QC-FND-010	Authorship	Low	QC-ERR-010	UR-WEB-08 and UR-WEB-09 use lowercase "the portal shall" inconsistent with all other requirements.
QC-FND-011	Authorship	Low	QC-ERR-011	SRS sign-off section contains entirely blank approver signatures and dates for all four designated approvers.
QC-FND-012	Omission	Medium	QC-ERR-012	RBAC used extensively throughout but not defined in Appendix A Glossary.

Table D-1: QC Findings Summary

References

- BioSec Educational Institution, *Project Specification: Biometric Student Attendance System* (BS-PS-2026-001, Version 1.0), ICT2113, 2026.
- Nevon Solutions, *Software Requirements Specification (SRS): Biometric Student Attendance System* (NS-SRS-2026-001, Version 0.8), 3 March 2026.
- Kevin Anthony Jones, *Labtorial Guide for Class Project: Reqrs now & near future*, Dec 2025.
- Kevin Anthony Jones, *Rubric for Assessment of Validating Report*, Mar 2026.
- Karl E. Wiegers, *Software Requirements Specification Template*, 1999.